# Read the Room: Grasp-Aware Planning and Learned Control for Multi-Robot Transport

Anonymous Author(s)

***Abstract*— Shared policy weights do not determine how robots coordinate through a common load. We study the planning and execution interfaces of two wheeled bimanual carriers using grasp-conditioned assembly geometry and shared learned local control. Assigned grasps and attained carrier placement define aperture constraints and object-level MPC references; a supervisor selects lift and transport phases. Including the carriers narrows a recorded beam's admissible yaw from $25.8^\circ$ to $5.5^\circ$. A frozen ten-scene simulation test with independent activation delays completes 35/40 missions; all failures precede verified-route execution. A later twelve-scene cohort completes 46/48 with a different actor and grasp configuration. Fixed-grasp spatial probes extend the reference to object roll within a $90^\circ$ operating envelope: a geometric teacher reaches $44.29^\circ$ and $88.29^\circ$ for $45^\circ$ and $90^\circ$ targets, while the separate learned actor fails its $45^\circ$ probe. These results separate geometric feasibility, supervised coordination and controller limitations. Training centrally pools teacher-labeled local observations; execution shares object references and readiness through phase/height commands. The evidence characterizes simulated transport under welded retention, without establishing a learning advantage or minimum communication.**

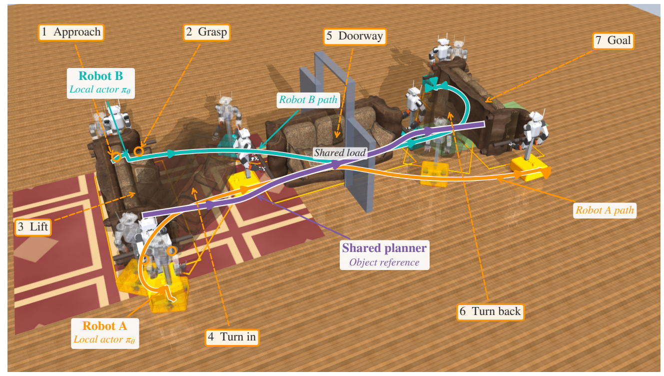


Fig. 1. Recorded cooperative transport (seed 142000). Both robots execute the same local policy with references from one shared high-level planner; a supervisor selects phase and height. Seven numbered stages show approach, grasp, lift and doorway reorientation. Orange/Tiffany-blue curves are measured Robot A/Robot B base paths; purple is the planned object path. Curves are projected onto the floor and overlaid for visibility. Translucent poses expose intermediate motion; orange rings mark grasp locations and cutaway walls show the doorway.

## I. Introduction

Can scaling a single-robot policy also produce a collaborating team? Larger datasets and shared models broaden the tasks and embodiments served by one policy [1]. Yet when copies run on different robots, will their actions complement one another, or must they communicate what they intend to do? Shared weights provide a common action rule; different observations and physical states still leave the team to coordinate intent, timing and support. Recent learned carrying and team policies make this distinction concrete [2]–[4].

Consider two robots carrying a couch through a narrow doorway. They must reach separated grasps, acquire contacts and lift together before turning through the opening. A couch that fits may leave insufficient room for its carriers. Their attained layout can differ from the nominal grasp arrangement, and their articulated arms allow that layout to change during motion. The challenge therefore spans acquisition and navigation: a feasible object path must also accommodate the team that executes it.

Self-driving offers a useful comparison. Decentralized driving policies choose actions for individual vehicles while accounting for other road users [5]. Their decisions interact, but their bodies are ordinarily mechanically separate. Robots holding a couch share a dynamical system: one robot's acceleration changes the other's load, and one robot's lift moves the other's grasp target. Local action selection must therefore remain compatible with both the intended motion and changing physical support.

Human collaboration suggests a possible interface. Partners can agree to "turn through the doorway" while adjusting their movements through vision and touch. Haptic feedback can improve coordination [6]; task-level agreement accompanies continuous physical feedback. For robots, this motivates sharing goals, grasp roles and motion constraints while each robot determines its own base, arm and lift actions. How much additional information is needed when a partner arrives late or changes support remains an open question.

We study *grasp-conditioned shared motion*: the task specifies where the object should go, while the attained carrier layout determines which motions are admissible. This differs from following an inferred partner wrench, as in PAINT [7]. The central interface maps assigned grasps and post-lift base placement to an assembly footprint, aperture-aware search and object-level MPC. Identical local actors track the resulting reference; a supervisor supplies phase and height. A separate spatial interface commands fixed-grasp roll within $90^\circ$. The contribution is the connection between acquired geometry, shared motion and execution conditions, rather than a new intent estimator or locomotion backbone.

Our contributions are:

- **Grasp-conditioned geometric feasibility**: an attained-formation aperture model and a displacement bound linking carrier drift to clearance.
- **A shared-reference control formulation**: common object motion and supervised progression drive identical local actors, with a separate fixed-grasp spatial reference and an explicit $90^\circ$ roll envelope.

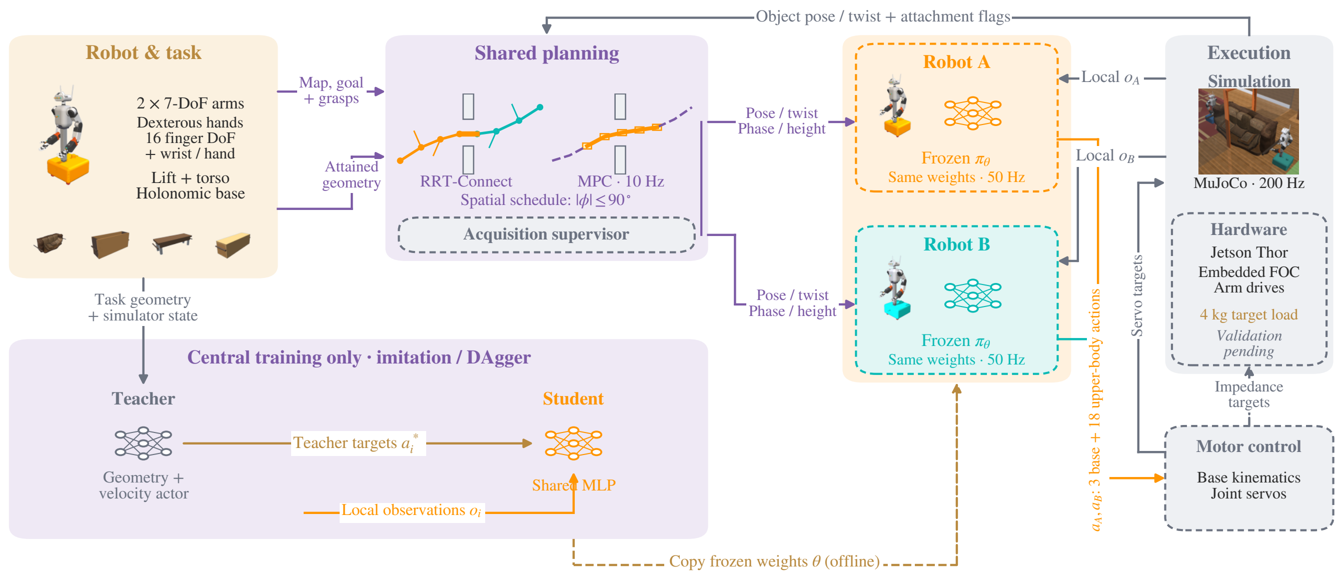


Fig. 2. Shared planning and local execution. Signal names are placed on arrows; blocks identify computations. Shared pose/twist, phase and height references reach both frozen actors, while $o_i$ contains only that robot's joint/hand feedback and object-relative observations. Central imitation/DAgger uses teacher action labels; dashed weight copying occurs offline. Motor conversion supplies servo targets in simulation and impedance targets to the hardware interface. Hardware system validation remains pending.

- **An evaluation across acquisition, transport and reorientation**: frozen 40- and 48-mission cohorts, recorded spatial probes and force-response diagnostics, with controller and contact-model differences retained.

The experiments use idealized contact-gated welded retention and test new seeds within the training scene families. They do not establish superiority over a matched geometric controller, generalization to unseen layouts, or minimum communication. Communication amount, content and task-level agreement are discussed as subsequent research questions.

## II. Related Work

*a) Physical interaction and inferred intent:* Cooperative transport can be organized around following a partner's guidance or tracking an agreed object motion. In the former setting, physical interaction carries information about the intended motion: role-based learning treats a partner's actions as implicit communication [8], and haptic feedback contributes to human coordination [6]. The observation interface determines how that information reaches the controller. PAINT [7] represents intent as planar force and yaw torque. Its privileged teacher observes the interaction wrench; a student estimates it from arm proprioceptive history and learns with wrench regression and KL-regularized teacher guidance. A frozen locomotion policy executes the resulting base commands. This separates intent inference from locomotion and supports deployment without explicit payload tracking. Our actors instead receive object state and a shared geometric reference; we do not estimate a partner-applied wrench. The complementary question is how the attained grasp and carrier arrangement constrain the motion that a team can jointly track.

*b) Shared policies and contact coordination:* Broad individual skills provide a foundation for teamwork. TidyBot++ supports mobile-manipulation demonstration collection [9], while CrossFormer unifies heterogeneous observations, actions and control rates [1]. CHORUS adapts a shared VLA to concurrent collaboration using local visual observations and robot identity/role prompts [4]. These approaches make observation and timing part of the coordination problem even when weights are shared. Physical contact further constrains the interface: decMBC conditions local control on the geometry of rigidly attached carriers [2]; decPLM maintains surface contact through a learned arm/base hierarchy and constellation rewards [3]. Its initial-only versus continuous contact-frame comparison links information requirements to payload shape and load. Sequential Asymmetric Imitation trains against progressively more realistic partner behavior, connecting interaction experience to learned waiting and yielding [10]. Our frozen evaluations retain an explicit acquisition supervisor; they do not establish the same local readiness or partner-inference capability.

*c) Geometry and explicit shared references:* Paired-grasp planning couples grasp selection to cooperative transport [11], while semi-coupled planning supplies a common object path with local redundancy resolution [12]. Distributed MPC exchanges predictions to coordinate coupled motion [13]; learned encodings compress such messages while retaining trajectory reconstruction [14]. A task goal, an object reference and a predicted partner trajectory therefore provide different information. Our formulation fixes grasp assignments, measures carrier offsets after lift, and uses their assembly footprint for aperture-constrained planning and object-level MPC. A displacement bound relates formation drift to reduced-model clearance, while separate spatial probes expose the limits of fixed-grasp roll. This focuses the contribution on the geometry and progression of a shared reference; a matched study is still needed to determine how much of that reference can be removed.

## III. Grasp-Aware Planning

The task specifies an object, assigned hand–rail contacts, a static map and a goal pose. Two wheeled bimanual robots acquire and lift the object before planning its transport. The attained carrier layout defines a planar assembly model for global search; object-level MPC supplies a common motion reference to the local actors (Fig. 2).

### *A. Grasp geometry as a team constraint*

Let $q = (t, \theta) \in \mathrm{SE}(2)$, $t = (x, y)$, denote object pose and $s_i = (u_i, v_i)$ the nominal position of base $i$ in the object frame. With payload rectangle $\mathcal{P}$ of dimensions $(L, D)$ and base-disc radii $r_i$,

$$\mathcal{C}(s) = \mathcal{P} \cup \bigcup_{i=1}^{2} B(s_i, r_i), \qquad \mathcal{C}_q = t + R(\theta)\mathcal{C}(s). \tag{1}$$

Calibrated object profiles specify reachable grasps. Base offsets $s_i$ are measured after lift and held fixed for route planning. This is a formation approximation: articulated arms allow the physical bases to move relative to the object. For obstacles $\mathcal{O}$ and signed clearance $d$, the configuration-space problem [15] is

$$\begin{aligned} &\gamma : [0,1] \to \mathrm{SE}(2), \quad \gamma(0) = q_{\text{lift}}, \quad \gamma(1) = q_g, \\ &d\big(\mathcal{C}_{\gamma(\tau)}, \mathcal{O}\big) \ge \varepsilon \quad \text{for all } \tau \in [0,1]. \end{aligned} \tag{2}$$

The collision model excludes supports below a specified carrying-height bound and represents the payload and bases, not the articulated swept volume.

The sensitivity to formation drift can be stated without assuming rigid attachment of the bases. Let $s^0$ denote planning offsets and $\delta_s(t) = \max_i \|s_i(t) - s_i^0\|$. At the same object pose $q$, translation of each base disc by at most $\delta_s$ gives

$$\mathrm{dist}(\mathcal{C}_q(s(t)), \mathcal{O}) \ge \mathrm{dist}(\mathcal{C}_q(s^0), \mathcal{O}) - \delta_s(t). \tag{3}$$

The triangle inequality gives this reduced-footprint bound. Clearance above $\delta_s$ suffices at the same object pose; articulated swept volume and path error are excluded. The test is retrospective, not enforced online.

For a centered opening of width $W$, define $w_{\mathcal{P}} = L|\sin\theta| + D|\cos\theta|$ and

$$\begin{aligned} w_i(\theta; s) &= 2\big(|u_i \sin\theta + v_i \cos\theta| + r_i\big), \\ \bar{w}(\theta; s) &= \max\{w_{\mathcal{P}}, w_1(\theta; s), w_2(\theta; s)\}. \end{aligned} \tag{4}$$

The aperture condition $\bar{w} \le W - 2\varepsilon$ is exact for mirrored, equal-radius layouts and conservative otherwise. Yaw scanning and boundary bisection identify admissible intervals.

The corresponding gate-pose set couples orientation and lateral placement:

$$\mathcal{G} = \left\{ (x_d, y, \theta) : |y - y_d| \le \frac{W - \bar{w}(\theta; s)}{2} - \varepsilon \right\}. \tag{5}$$

Here $(x_d, y_d)$ is the opening center; a negative bound excludes that yaw. This sampling constraint does not certify clearance elsewhere. For the recorded beam handoff, the object-only admissible yaw window is $25.8^\circ$; including the carriers reduces it to $5.5^\circ$ (supplementary geometric illustration).

Gate-biased RRT-Connect [16] samples the opposite root, aperture set and workspace with probabilities $(0.10, 0.35, 0.55)$. Steering uses $d_q = \|t_a - t_b\|_2 + 0.6|\,\mathrm{wrap}(\theta_a - \theta_b)|$ and 0.35 m-equivalent extensions. Search permits three seeded attempts of 20,000 iterations, followed by shortcutting and clearance-checked smoothing. Swept collision checks use 5 mm increments and a 22 mm margin; an independent 10 mm-resampling verifier requires $\varepsilon = 20$ mm along the delivered path, including joins. These finite-resolution checks implement Eq. (2); failed verification rejects the route.

### *B. Shared receding-horizon object reference*

Object MPC [17] predicts $q_k$ under object-frame twist $\nu_k = (v_x, v_y, \omega)$ with timestep $h$:

$$q_{k+1} = f(q_k, \nu_k) = q_k + h\ \mathrm{diag}(R(\theta_k), 1)\nu_k. \tag{6}$$

At each solve, measured object pose and base offsets update the model. Offsets are parameters, not control decisions. Monotone measured path progress selects $\bar{q}_{1:N}$, so elapsed time alone cannot advance the reference during a stall. With $e_k = q_k - \bar{q}_k$ and locally unwrapped yaw, successive convexification solves

$$\begin{aligned} \min_{q_{1:N}, \nu_{0:N-1}, \sigma \ge 0} \quad & \textstyle\sum_{k=1}^{N-1} \|e_k\|_Q^2 + \|e_N\|_{Q_f}^2 \\ & + \textstyle\sum_{k=0}^{N-1} \|\nu_k\|_{R_\nu}^2 + \lambda \|\sigma\|_1 \\ \text{s.t.} \quad & q_0 = \hat{q}, \quad q_{k+1} = A_k q_k + B_k \nu_k + c_k, \\ & a_{jk}^\top q_k + \sigma_{jk} \ge b_{jk} + \varepsilon, \\ & |\nu_k| \le \nu_{\max}, \quad |\nu_k - \nu_{k-1}| \le h\dot{\nu}_{\max}, \\ & \|\nu_{xy,k} + \omega_k J \hat{s}_i\|_\infty \le 1.2 \text{ m/s}. \end{aligned} \tag{7}$$

Here $J$ rotates by $90^\circ$ and $\hat{s}_i$ is the measured base offset. Affine dynamics approximate Eq. (6); separating planes approximate obstacle clearance with individual nonnegative slacks $\sigma_{jk}$. The frozen profile uses $N = 20$, $h = 0.1$ s, $Q = \mathrm{diag}(12, 12, 30)$, $Q_f = 40Q$, $R_\nu = \mathrm{diag}(0.6, 0.6, 0.8)$ and $\lambda = 4000$. Two warm-started convexification passes use OSQP [18]. Limits are 0.12 m/s per translation axis and 0.10 rad/s yaw, with slew bounds of 0.15 m/s$^2$ and 0.10 rad/s$^2$. The first control is accepted only if the solve succeeds, the command is finite, and maximum slack is at most 10 mm; otherwise the bridge commands zero twist. Execution clearance is measured independently of the QP slacks.

For world-frame station offset $\rho_i$ from the object center, the common reference induces rigid-motion feedforward

$$v_i^{\text{ff}} = R(\theta)(v_x, v_y)^\top + \omega \hat{z} \times \rho_i, \qquad \omega_i^{\text{ff}} = \omega. \tag{8}$$

Each robot transforms this reference into its base frame and adds bounded learned residuals. Both planners use the same assembly geometry and requested clearance. Reference updates preserve the continuous physical state.

### *C. Fixed-grasp object reorientation*

Spatial pose tracking extends the reference interface; it does not replace planar RRT/MPC with a collision-free SE(3) search. Let $T_O = (R_O, p_O) \in \mathrm{SE}(3)$ and let $\bar{T}_{ih}$ be the fixed object-to-hand transform for robot $i$, hand $h \in \{L, R\}$. The reference satisfies

$$\begin{aligned} T^*_{ih}(t) &= T^*_O(t)\bar{T}_{ih}, \\ R^*_O(t) &= R_{O,0} \exp\big(\phi(t)[e_x]_\times\big). \end{aligned} \tag{9}$$

where $R_{O,0}$ is the lifted orientation and $e_x$ the object's long axis. Upper/lower handles provide distinct hand targets during roll. We adopt the fixed-grasp physical operating envelope

$$|\phi(t)| \le \phi_{\max} = \pi/2, \qquad \bar{T}_{ih}(t) = \bar{T}_{ih}(0). \tag{10}$$

Beyond 90°, continuing the same hand assignment twists the arms and requires switching hands along the handles. Such contact reassignment and regrasp planning are outside this paper. This mechanism-specific envelope is not a guarantee of reachability for every pose within it. It bounds object roll relative to the lifted frame, not world-frame yaw during transport. Together, the planar and spatial tests examine cooperative translation, yaw and out-of-plane rotation (Fig. 3); independent pitch-axis control is not evaluated.

The high-level reference specifies a clearance lift followed by roll. Over a segment beginning at $t_a$ with duration $\tau$, set $u = (t - t_a)/\tau \in [0, 1]$ and interpolate

$$\begin{aligned} s(u) &= 10u^3 - 15u^4 + 6u^5, \\ p^*(u) &= p_a + s(u)(p_b - p_a), \\ R^*(u) &= \exp\big(s(u)\log(R_b R_a^\top)\big) R_a. \end{aligned} \tag{11}$$

Differentiation supplies the centroid twist. Unlike planar MPC's measured path progress, these pose probes use a timed schedule after supervised lift. The local actor receives pose/twist errors and hand targets in its base frame; the geometric teacher instead maps hand errors to bounded Jacobian rates with joint-limit and selected self-clearance constraints. Neither controller chooses the requested endpoint. Large commanded roll replaces the planar upright-object guard with pose-tracking and contact guards; the 2° base-inclination limit remains.

## IV. Local Control and Coordination

### A. Robot platform and motor interface

Each carrier has a holonomic three-module steering/drive base, a two-stage lift, a pitching torso, two 7-DoF arms and two hands with 16 finger DoFs and an additional wrist each (Fig. 2A). The planar policy commands three base and eighteen upper-body coordinates; hand articulation uses stage-dependent closure rather than learned finger-level actions. The hardware interface uses Jetson Thor for local inference and CAN arm-drive commands $(q^d, \dot{q}^d, K_p, K_d, \tau^{\mathrm{ff}})$:

$$\tau^d = \mathrm{clip}\big(K_p(q^d - q) + K_d(\dot{q}^d - \dot{q}) + \tau^{\mathrm{ff}},\ -\tau_{\max}, \tau_{\max}\big). \tag{12}$$

Embedded field-oriented control (FOC) regulates arm-motor current; MuJoCo uses its actuator model instead. The 4 kg hardware target load is distinct from the evaluated 1–2 kg simulation loads. Hardware system performance and that load target are not validated by the reported experiments.

### B. Shared policy and local observations

In the frozen planar cohort, each robot executes $a_i = \pi_\theta(o_i)$ at 50 Hz with identical weights; MuJoCo dynamics run at 200 Hz [19]. The 157-dimensional observation includes joint state and tracking errors, base motion and gravity, object-relative pose/twist, grasp geometry, palm-target and docking errors, phase/height references, previous action and commanded motion. It excludes partner joint/velocity streams. Simulation supplies object state and six clipped constraint-multiplier channels for grasp feedback.

For base position $p_i$ and yaw $\psi_i$, relative features are

$$\begin{aligned} \tilde{p}_i &= R(-\psi_i)(t - p_i), \qquad \tilde{s}_i = R(\theta - \psi_i)s_i, \\ c_i &= \big(\cos 2(\theta - \psi_i),\ \sin 2(\theta - \psi_i)\big). \end{aligned} \tag{13}$$

The double-angle feature represents footprint symmetry; assigned hand targets retain contact identity. This common coordinate convention permits weight sharing between opposed carriers without assuming policy equivariance. The 512–256–128 ELU MLP produces three base and 18 upper-body actions. Arm targets are posture residuals; lift actions specify a bounded 0.06 m/s rate during lifting and position residuals around the attained carrying posture during transport.

### C. Acquisition and asynchronous progression

Assigned rails define the approach targets. The geometric teacher stages, docks and refines wrist pose within 40 mm. Panel/beam profiles use 0.18 m hand spacing instead of 0.40 m and reserve 60 mm downward lift travel. Robot $i$ attaches when both assigned palms are within 35 mm and 30° of their targets, base speed is below 0.10 m/s, and both palm/finger contacts apply more than 0.1 N compression to the correct rails. Weld constraints then retain the attained transforms [20]; subsequent frictional slip is not modeled.

One actor starts immediately and the other after $\Delta \in [0, 8]$ s; the delayed robot holds its initial command until activation. Let $b_i$ indicate attachment and $z_0$ initial object height. The attached robot's height reference is

$$z_i^* = \begin{cases} z_0 + 0.04 \text{ m}, & b_i = 1,\ b_1 b_2 = 0, \\ z_0 + 0.12 \text{ m}, & b_1 b_2 = 1. \end{cases} \tag{14}$$

During early lift, the base holds and upward action is clipped at a 40 mm centroid rise. Remaining support permits this partial lift while the late robot tracks its moving grasp targets. Joint attachment releases the full lift. Transport begins after height error remains below 20 mm and object roll/pitch below 0.06 rad for 25 control steps (0.5 s). The same actor then tracks MPC references without resetting the physical state. Gated-start comparisons instead synchronize controller availability and attachment.

### D. Information and physical assumptions

Let $x_t$ be the simulator state, $b_{i,t}$ attachment status, and $m_t = (\mathcal{R}_t, \kappa_t, z_t^*)$ the shared reference: motion, phase and height. The evaluated information flow is

$$\begin{aligned} m_t &= \mathcal{H}(x_{O,t}, b_{1,t}, b_{2,t}, g), \\ o_{i,t} &= f_i(x_t, m_t), \qquad a_{i,t} = \pi_\theta(o_{i,t}), \\ \mathcal{D} &= \bigcup_{i,t} \{(o_{i,t}, a^*_{i,t})\}, \\ \theta^* &= \arg\min_\theta \sum_{(o,a^*)\in\mathcal{D}} \|\pi_\theta(o) - a^*\|^2_W. \end{aligned} \tag{15}$$

Here $g$ is task geometry, $\mathcal{R}_t$ the motion reference and $W$ a diagonal action-weight matrix. Central training pools teacher-labeled samples from both robots; teacher labels may use simulator geometry and supervised progression. Student execution retains identical weights and normalization, but no teacher labels, optimizer state or partner joint/velocity streams.

Shared memory delivers simulator object state, 10 Hz planar MPC references and supervisor-selected phase/height to the 50 Hz actors. Spatial probes sample the common pose schedule at the control rate. These references convey joint readiness even without direct actor-to-actor messages. Configured update rates are not measured network traffic. The separate PPO and local-history studies use centralized critics during training only; their information and results are specified in the supplement.

The frozen planar actor's six clipped feedback channels aggregate equality constraint multipliers, not calibrated wrist wrenches. Welds can sustain tensile forces and moments without frictional retention limits. The supplement defines the feature scaling and physical assumptions; no load-capacity or actuator-saturation claim is made.

## V. Experimental Evaluation

### A. Evaluation scope

The original frozen cohort uses four held-out seeds per scene within the ten collection scene families and one trained actor. Later planar, spatial and local-history evaluations retain separate checkpoints and denominators. Neither unseen-family generalization nor training-seed robustness is tested.

### B. Training protocol

The planar actor uses imitation and DAgger [21]: a geometric teacher labels acquisition/lift and a saved velocity actor labels transport. Three rounds collect 30 episodes each, first with the teacher and then with the student under teacher labeling. The 524,490-sample dataset includes 345,484 imported development samples; 86/90 collection episodes complete stable holding. Training centrally minimizes phase-balanced weighted action MSE across both robots. The actor receives no subsequent PPO update.

Collection randomizes mass $U[1, 2]$ kg, dimensions $U[0.95, 1.05]$, rail friction $U[0.9, 1.8]$, base offsets 0.08–0.50 m, headings $\pm 20^\circ$ and activation delays 0–8 s. Automatic curricula are disabled. Episodes truncate at 35 s or stop upon physical failure or completed collection. Furnished-scene collection holds after lift; RRT/MPC supplies navigation at test time. The supplement specifies imported-data provenance, feature rebasing, normalization, loss weights and fitting settings.

### C. Mission test and outcome criteria

The frozen test comprises four prespecified seeds in each of ten scenes (Table I): nine doorway environments and an open arena. Each mission starts detached and has a 180 s simulation budget. Mass, dimensions, base offsets, headings and activation delays use the DAgger ranges above; object XY and yaw additionally vary by $\pm 0.10$ m and $\pm 0.20$ rad. Thus visually large payloads represent 1–2 kg loads. The paired MPC tests use a separate actor and domain (Table II); their results are not pooled.

TABLE I
Scene geometry and frozen outcomes: four trials per scene.

| Scene | Object | $L \times D \times H$ (m) | $S$ (%) | $T_L$ | $T_G$ |
|---|---|---|---|---|---|
| Living | Sofa | $2.73 \times 1.12 \times 0.93$ | 75 | 9.0 | 98.3 |
| Bedroom | Wardrobe | $2.10 \times 0.62 \times 0.95$ | 100 | 11.7 | 90.9 |
| Dining | Table | $2.40 \times 0.76 \times 0.75$ | 100 | 8.6 | 94.9 |
| Dock | Case | $2.80 \times 0.82 \times 0.80$ | 100 | 9.3 | 106.3 |
| Warehouse | Crate | $2.20 \times 0.70 \times 0.65$ | 100 | 9.7 | 96.4 |
| Gallery | Panel | $1.90 \times 0.09 \times 1.25$ | 50 | 11.6 | 93.1 |
| Workshop | Beam | $3.20 \times 0.20 \times 0.20$ | 75 | 15.0 | 122.1 |
| Office | Table | $2.40 \times 1.10 \times 0.74$ | 75 | 9.1 | 93.5 |
| Apartment | Sofa | $2.10 \times 0.86 \times 0.78$ | 100 | 10.7 | 91.8 |
| Arena | Sofa | $2.10 \times 0.90 \times 0.83$ | 100 | 11.3 | 40.3 |
| All (40) | | – | 87.5 | 10.6 | 95.4 |

Nominal dimensions; variation $\pm 5\%$. $T_L/T_G$: median seconds to completed lift/first goal among passes. Simulation time; planning compute is excluded.

TABLE II
Evaluation protocols and original denominators.

| Test | Domain / comparison | Pass |
|---|---|---|
| Frozen missions | 10 scenes; 1–2 kg; independent starts, $\Delta \leq 8$ s; 180 s. | 35/40 |
| Original MPC | 8 scenes; 1.5 kg; gated starts, $\Delta \leq 0.8$ s; 180 s. | 36/40 |
| Aligned MPC | Same actor and 40 seeds as original MPC; geometry and margin change. | 40/40 |

Let $\mathsf{F}$ denote physical termination at any point in a mission. Failure guards reject object heights outside $[z_{\text{rest}} - 0.12, z_{\text{rest}} + 0.30]$ m, object roll/pitch exceeding 0.6 rad, base inclination above $2^\circ$, or forbidden world contact. Base inclination is checked every physics substep. With final object position $t_T$, yaw error $\delta\theta_T$, completed-lift flag $\ell$, attachment flags $b_i$, and minimum measured assembly clearance $d_{\min}$, mission success is

$$S = \mathbf{1}\{b_1 b_2 = 1,\ \ell = 1,\ \|t_T - t_g\| < 0.10,\ |\delta\theta_T| < 0.10,\ d_{\min} \geq 0,\ \neg\mathsf{F}\}. \tag{16}$$

Position and angle tolerances are in meters and radians. Incomplete acquisition/lift and rejected routes count as failures. Planned clearance (20 mm) and measured execution clearance (0 mm) are separate criteria. All original trials remain in the denominator; Fig. 3 replays recorded states. Computation time is reported separately from simulation duration.

### D. Geometric compatibility and asynchronous acquisition

The paired development test changes the local disc cover and 30 mm margin to the global rectangle model and 20 mm margin. Completion rises from 36/40 to 40/40, recovering four living-room stalls with the same actor (supplementary planning trace). Geometry, margin and archived implementation settings change together; this comparison diagnoses incompatibility rather than isolating a planning improvement.

With a 7.68 s activation gap, attachment occurs at 1.88/10.76 s and transport starts at 13.44 s (supplementary trace). Early lift accommodates a late partner through su-

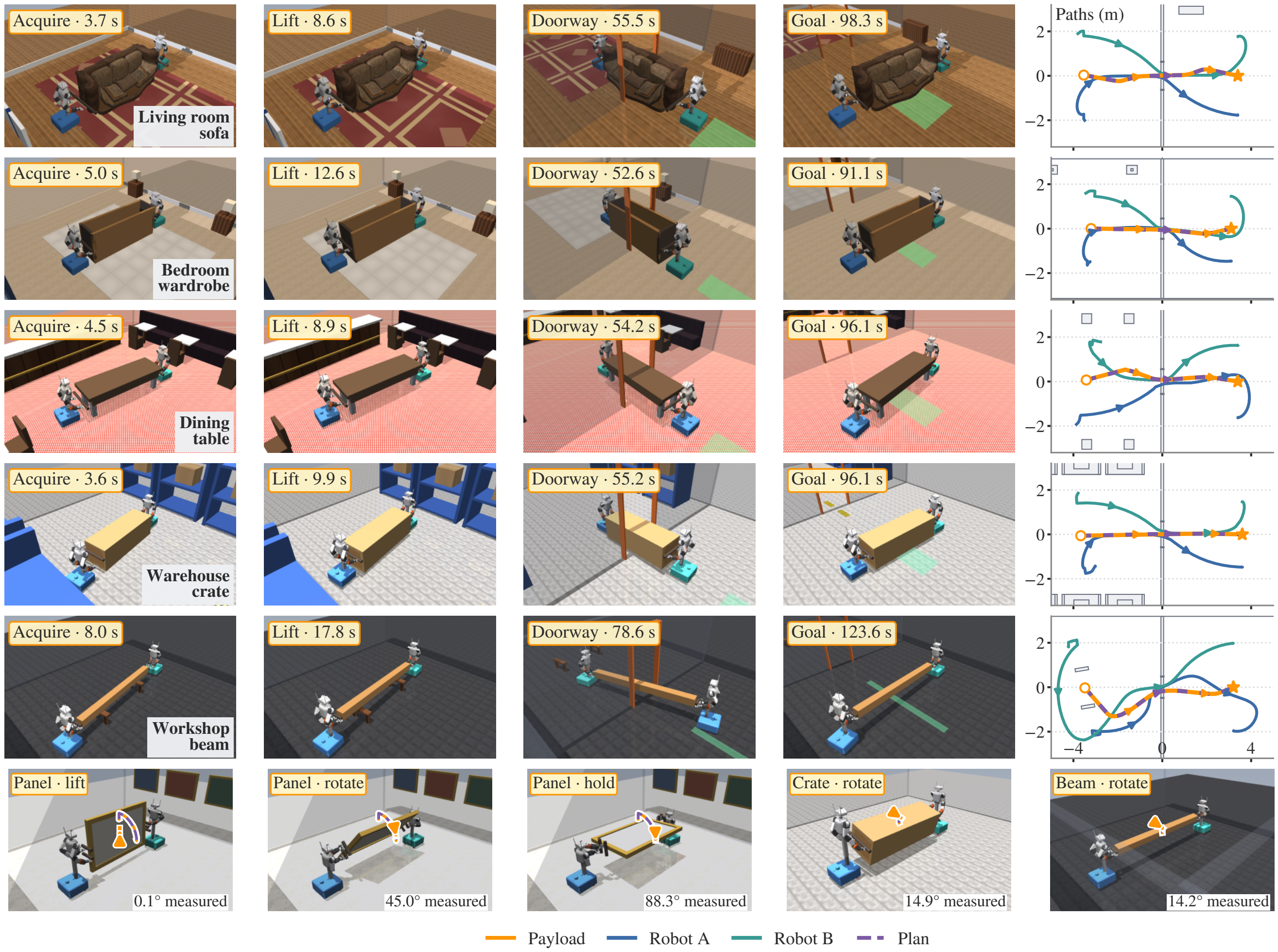


Fig. 3. Cooperative object-pose execution. Top five rows retain the original sofa, wardrobe, table, crate and beam missions: acquisition, lift, doorway passage, arrival and planar trajectories. Both position and yaw vary. Bottom row adds teacher-controlled out-of-plane reorientation: panel lift, intermediate/terminal roll, and single-bar crate/beam roll. Arrows show motion direction; purple dashed/orange solid paths are commanded/measured. Reorientation paths track a representative object-fixed point and labels give measured roll. Robot bodies retain their original materials; blue/teal bases identify Robot A/B. These are separate trials, not a combined tilt-through-door experiment. Table I retains all frozen mission outcomes.

pervised progression; the delayed beam failure shows that acquisition geometry can still prevent completion.

A same-start panel replay (seed 142020) succeeds with the geometric teacher where the frozen actor tips the panel. This diagnostic motivates a matched full-cohort baseline; it does not estimate overall teacher performance. The separate pickup PPO study and its reward equations are supplementary.

### E. Reliability across acquisition and transport

The original frozen test completes **35/40 missions (87.5%)**. Failures comprise two panel tip-overs before grasp, a delayed beam acquisition timeout, an incomplete office-table lift and an exhausted living-room route search. Of 40 starts, 37 achieve joint attachment, 36 complete lift, and 35 obtain and execute a verified route (supplementary stage counts). The conditional 35/35 transport result applies to successfully acquired states; acquisition remains part of team reliability. Four trials per scene support descriptive counts, not precise scene-level reliability estimates.

For 31 successful doorway missions, median planned/measured minimum clearances are 32.5/29.0 mm, with a median within-trial difference of 2.2 mm. One reaches 19.95 mm: above the execution threshold but below the planning margin. These measurements concern control-step payload/base footprints, not continuous clearance of the articulated robot geometry. Among successful missions, median first goal arrival is 95.4 s and completed lift 10.6 s; transport accounts for a median 88.6% of duration (supplementary duration analysis).

Median search and smoothing times are 5.15 and 19.83 s; their per-mission sum has median 26.45 s (11.41–70.70 s), excluding verification and retries. These wall-clock costs are excluded from simulated completion times. Configured 10 Hz MPC and 50 Hz actor rates therefore do not establish real-time execution.

### F. Formation variation and operational timing

Across the 35 successful records, base-offset deviation from first completed lift has median per-trial maximum

16.2 mm and overall maximum 40.8 mm. The latter exceeds the nominal 20 mm margin, so Eq. (3) does not uniformly certify these executions. Measured clearance remains an empirical reduced-model check.

Global search and verification pause simulation after lift; active holding during planning is untested. The logs omit complete MPC solve-time and deadline-miss distributions. Zero twist after a rejected solve is a fallback, not a safe-recovery guarantee.

### G. Spatial reorientation and later evaluations

The panel probes use a 1.5 kg, $1.90 \times 0.09 \times 1.25$ m object, supervised acquisition and welded retention. A 0.15 m clearance lift precedes roll; reference durations are 22.5/45 s for $45°/90°$. The geometric teacher reaches $44.29°$ and $88.29°$, respectively (Fig. 3). Passing requires 3 s of terminal holding with position error below 25 mm, geodesic rotation error below $3°$, linear/angular speed below 0.04 m/s and 0.04 rad/s, retained contacts and no physical termination. Only positive roll is tested. These are selected configuration-specific probes, not a success-rate estimate across the envelope in Eq. (10).

The separate spatial imitation actor completes the $0°$ and $15°$ probes of a three-angle, same-start test; the $45°$ probe terminates on robot self-penetration at $17.26°$ measured roll. The successful $15°$ probe ends at $12.83°$ (supplementary rotation traces). Teacher and learned probes use different starts and handle spans (0.20 versus 0.16 m); they do not isolate a controller effect. A historical $180°$ teacher attempt fails at $51.40°$ on the tracking guard. It remains in the supplementary record as an out-of-envelope diagnostic, not proof of a sharp physical threshold at $90°$.

Two additional prespecified teacher probes use one rail per carrier (vertical rail separation zero): a crate and a beam, each 1.5 kg, commanded to $15°$ with a 15 s rotation segment after clearance lift. They complete at $14.92°$ and $14.15°$, respectively, under the same terminal tolerances. These two nominal probes demonstrate small-angle rotation with single-bar grasps; they do not identify a maximum safe angle or isolate the effect of rail geometry across different objects.

A later frozen planar cohort completes 46/48 missions across twelve scenes, using a different actor and upper/lower panel grasps. Both failures are acquisition timeouts. This broader cohort retains supervised progression and welds; its checkpoint and domain differ from the 35/40 cohort, so the scores are not a matched improvement estimate. The supplement reports its full scene counts and the completed local-history recovery comparisons. Neither cohort evaluates large-angle rotation during doorway passage.

### H. From object commands to hand reactions

A separate instrumented teacher probe repeats the nominal $90°$ roll with a 1.5 kg panel (seed 207000). The planner commands pose and twist, not hand wrenches. For hand $h$ of robot $i$, selected weld multipliers $\lambda_{ih}$ give the payload

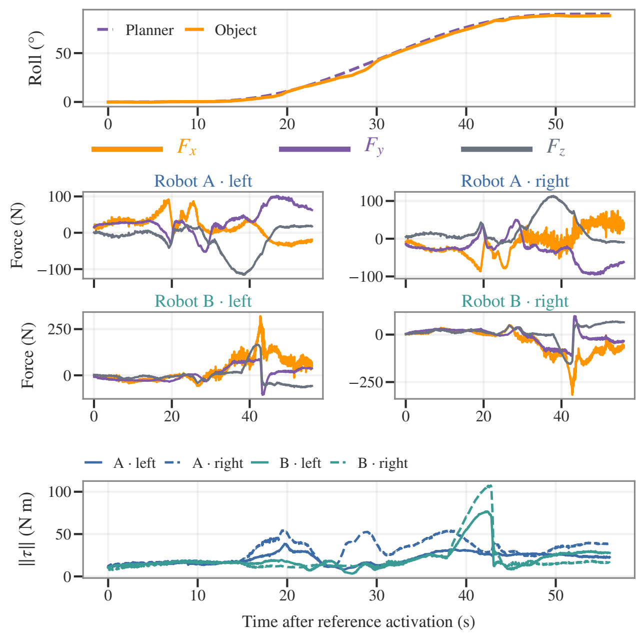


Fig. 4. Planner reference, measured object motion and per-hand reactions. The same pose-to-wrench interface applies to each hand; this diagnostic uses one instrumented $90°$ panel trial with the geometric teacher. Force components use the fixed lifted-object axes; moment norms are palm-centered. Dashed roll is commanded, while hand loads are measured weld reactions rather than force targets. All panels share the time axis; force scales vary. Full moment components are supplementary.

wrench through

$$J_O^\top w_{ih}^{O,W} = S_O J_{ih}^\top \lambda_{ih},$$

$$w_{ih} = \begin{bmatrix} R_0^\top F_{ih}^W \\ R_0^\top [\tau_{ih}^{O,W} - (p_{ih}^W - p_O^W) \times F_{ih}^W] \end{bmatrix}. \qquad (17)$$

Here $S_O$ selects the free payload's six generalized coordinates, $J_O$ maps their velocities to its world twist, and $J_{ih}$ is the weld Jacobian. $R_0$ fixes the lifted-object axes; moments act about the measured palm center $p_{ih}$. These solver reactions act on the payload and exclude separate finger-contact forces.

The probe completes at $88.29°$ (Fig. 4). Over 2,802 control samples, left/right force-norm RMS values are 73.1/71.7 N for Robot A and 94.7/97.1 N for Robot B. Summed hand forces match the separately logged robot totals within $10^{-6}$ N. Opposing reactions can therefore be obscured by aggregate load feedback: one hand reaches 363.2 N and another 107.4 N m despite the light payload. Successful pose tracking alone does not establish feasible grip loads. These weld-supported loads motivate frictional retention and actuator-limit tests; this probe does not measure wrench tracking or learned intent inference.

## VI. Discussion

### A. Geometry, control and physical scope

The aperture analysis shows that assigned carrier placement restricts object-path feasibility. The formation bound identifies a second requirement: execution must preserve enough of the planned layout to retain clearance. Observed drift prevents a uniform certificate at the nominal 20 mm margin. A geometry-by-margin experiment and ordinary versus aperture-biased

RRT-Connect under equal budgets are still needed to isolate planning benefits.

The 35/40 and 46/48 cohorts establish operation of supervised stacks, not the advantage of learned control. A matched geometric-teacher baseline must hold starts, planner, supervisor and contact model fixed. Spatial probes sharpen this distinction: teacher-controlled $90^\circ$ rotation is demonstrated, whereas the learned $45^\circ$ probe fails. Upper/lower grasp geometry alone does not establish learned reorientation capability. Rolls beyond $90^\circ$ require hand reassignment to avoid arm twisting and are excluded.

Welded retention, light loads and simulator-derived feedback limit physical interpretation. Joint-limit and selected collision guards do not certify the full articulated swept volume. The four-hand diagnostic exposes large opposing weld reactions despite successful motion tracking. Frictional retention, actuator saturation and hardware sensing require separate validation. The newer breakable-grasp local-history trials in the supplement expose recovery failures; they neither replace the welded cohorts nor establish broad robustness. Evaluation within development scene families and one training seed per actor also limits claims of generalization.

### B. Shared intent and local feedback

The broader questions remain **RQ1**: how much must be shared beyond local object feedback; **RQ2**: which message content matters at a fixed budget; and **RQ3**: when task-level agreement supports local control under changing partner behavior. Central training can exploit joint state without providing it to executing actors. Conversely, local actors can receive joint readiness implicitly through explicit shared references. These are distinct information boundaries.

The supervised cohorts and preliminary local-history trials use different contact models, observations and tasks. They are not a communication ablation. Testing the questions requires matched message-content and rate sweeps, locally inferred versus supervised progression, and accounting for shared perception traffic as well as direct robot messages.

## VII. Conclusion

Shared weights leave a team to coordinate where it can move and when it can begin. Grasp-conditioned assembly planning and shared learned local control complete 35/40 frozen missions and 46/48 in a separate later cohort. Carrier-aware apertures and measured formation drift link common geometry to execution. Spatial probes further demonstrate teacher-controlled fixed-grasp roll up to $90^\circ$, while retaining the learned controller's failures. Together, these results define a supervised planning–control baseline; matched experiments remain necessary to establish when task-level intent suffices for reliable local coordination.

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